Element C

High Priority:

- Three segments of the box
    - The assistant curator of ectotherms at the Nashville Zoo, Mr. Hanna, stated that he wanted there to be three boxes (each 20 in x 20 in x 36 in) with a modular style system. This would encompass the lengths of the different Komodo dragons (babies, females, and males). If a dragon is six feet long then they only need to use two boxes. A modular system would allow for the three parts to be disassembled and stored away easily.

The boxes would be set up in a system like this with connection points on the sides that would allow for easy assembly and disassembly of the boxes. The boxes would all be symmetrical so the boxes can be arranged any way.

- Sturdy, sufficient, and comfortable handles

- There should be handles on all three segments of the box so that the box is easier to transport for the zookeepers before and after testing on the Komodo dragons. The handles should be sturdy while also being comfortable. If it is possible, a handle that folded down onto the sides would be a preference by our stakeholder, Mr. Hanna. There should be at least 2 handles per lateral side of the box, but more handles are generally better.

Example of a handlebar similar to what we need to create for the boxes (8 total: 6 on the sides and 2 on the ends)

- Have a guillotine-style locking gate that can't open from the inside
  - Our stakeholder heavily suggested that we have a gate that prevents the Komodo dragon from leaving the box while being operated on. He also suggested that we use a guillotine style gate where a panel (whatever material we decide to use) can slide into all three segments. This makes it easy to slide a gate into and out of any section of the box. We will also need to include a lock so the panel won't just slide open if the box tips over. Using this, the Komodo dragon will not be able to escape from the box until the panel is removed.

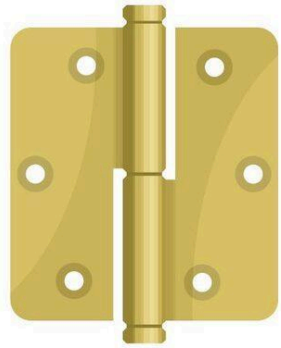
- Weight distribution/management
    - Our team is also on the task of weight management. The box needs to be able to be moved from storage and into the Komodo cage for testing. Our stakeholder explained that it would be much easier for the team if less people needed to carry the box. It will be our team's job to figure out what materials to use so that the box isn't too heavy even with the Komodo dragon in it (male Komodo dragons can weigh about 170-200 pounds).

- Visibility
    - When we visited the zoo, they also said that they were going to need to be able to watch the Komodo Dragon somehow. This means that we need to use clear plastic on at least some parts of the box. This is so the zookeeperes can watch the animal while they are doing medical testing on it. This would be an upgrade from their original product which is just a black plastic tube.

Medium Priority:

- Easy to clean
    - Our stakeholders would like to be able to easily clean the box. Though this is not a top priority, this is something that is very important. With the modular style box design, it should be easy to clean the box because each 3 ft x 2 ft x 2 ft box would be open on at least one side so someone can just clean the inside with what cleaning supplies they use.

- Folding the handles
    - Though adding handles is absolutely necessary, the handles' ability to fold in would be a bonus. Being able to fold the handles in would make the boxes easier to store in a smaller space. It would be useful to fold the handles down for storage and it would also be more comfortable if the handles fold all the way up for the people transporting the box.



In order to use folding handles, we would need to use some sort of hinge technique that would allow the handles to sway up and down easily.

Low Priority:

- Holding the tail/gate for only the tail
    - Holding the Komodo dragon's tail down would be useful for our stakeholders, but they don't require it. It would also be one of the hardest tasks to do because it would require strapping the tail every time the Komodo dragon needed testing and their tails are very strong (one whip from their tail has enough force to disable or kill prey as large as a deer).

This is an example of a strap we could use if we decide to try to help them strap the dragon down. This isn't a high priority because it doesn't have anything to do with the box. This would be almost like a separate problem to solve.

- Keepers must be able to lure dragons into contraption
    - There should be a little compartment in our box that allows for bait. The dragon's main lure is treats: small lizards, snakes, birds, and insects for younger dragons and rodents, monkeys, goats, wild boars, and deer for older ones. This shouldn't be the main focus of our project though because our stakeholder explained that they already have a bait that works efficiently.